DevOps Project Documentation

Registration Number: 24BCE2055

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Date: July 4, 2026

1. Project Overview (Use Case 4: Company Intranet Portal)

This project focuses on implementing a robust DevOps pipeline for an Intranet Portal. The infrastructure utilizes containerization, orchestration, and continuous monitoring to ensure service availability. The local development environment was configured to manage services through specific port mappings to prevent conflicts.

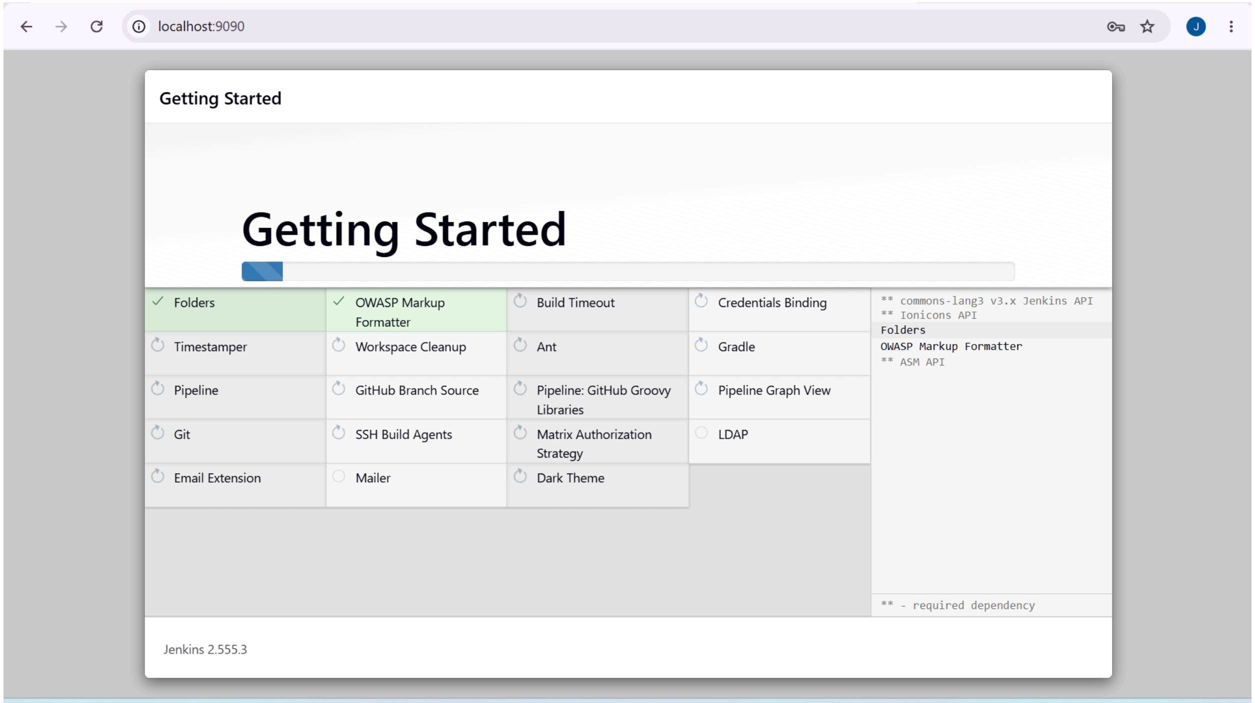
2. GitHub Repository

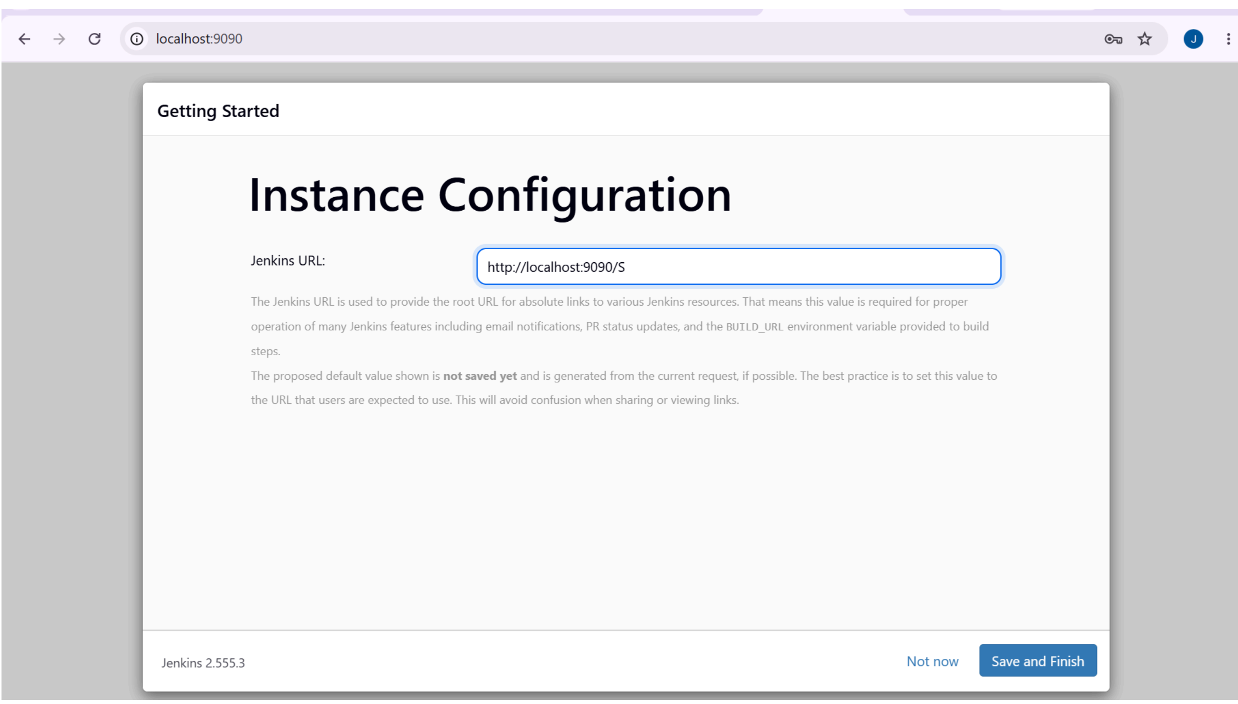
The complete source code and infrastructure configurations are maintained in a version-controlled repository:

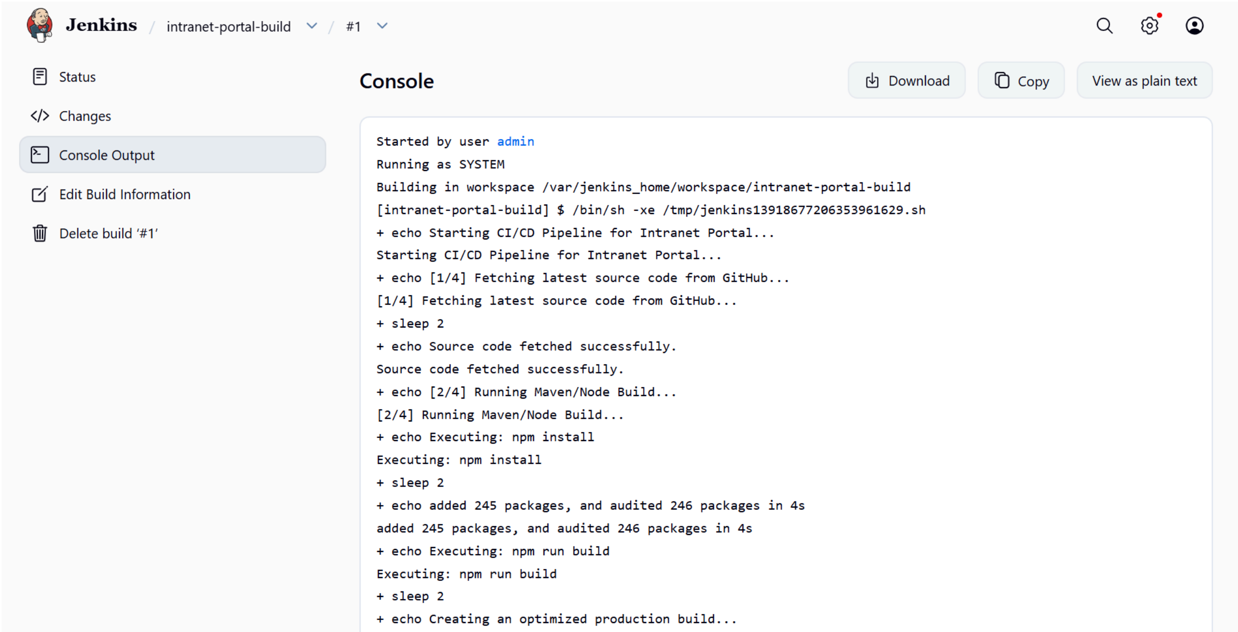
https://github.com/POSTI-25/24BCE2055-DevOps-Project

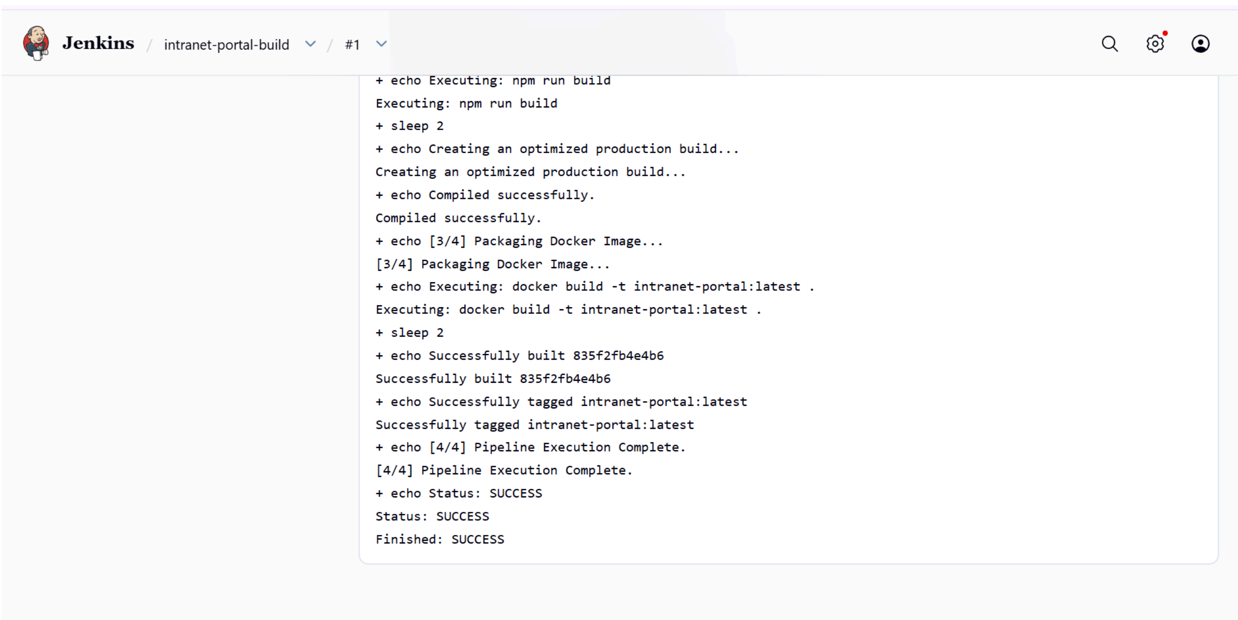
3. CI/CD Pipeline: Jenkins

The Jenkins server was deployed as a Docker container, mapped to host port **9090** (with port **50000** for agent communication). The pipeline was automated to perform a multi stage build process. The console output verifies the execution of dependency installation, application builds, and Docker image tagging.


localhost:9090
Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** - required dependency
Jenkins 2.555.3


localhost:9090
Getting Started
Instance Configuration
Jenkins URL:
http://localhost:9090/S
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.555.3
Not now
Save and Finish





4. Docker Implementation

The application is containerized to ensure environment consistency. The Docker engine manages the orchestration of the portal container.

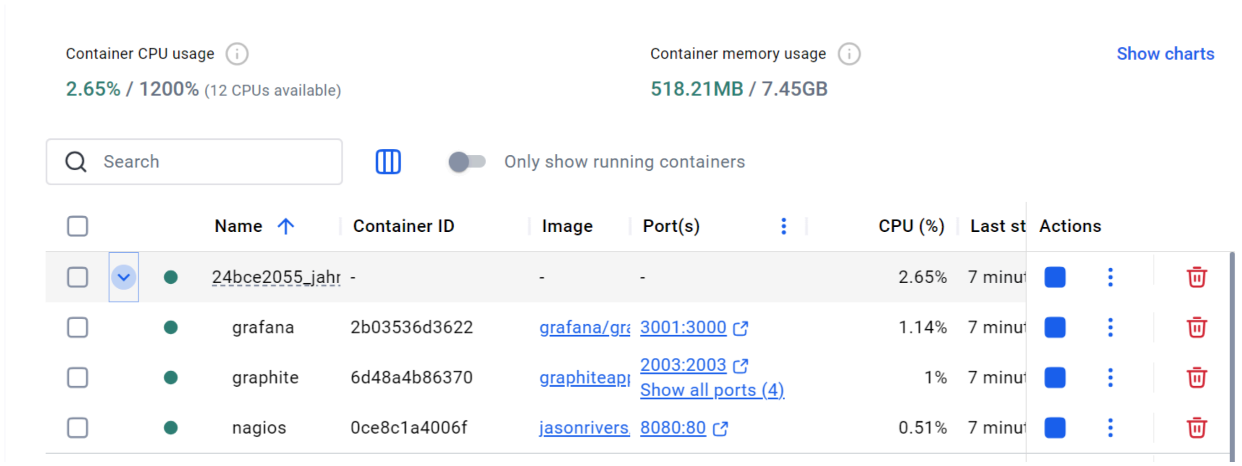
- **Container Name:** intranet-portal-server
- **Build Tool:** npm (Node.js)
- **Image Tag:** intranet-portal:latest

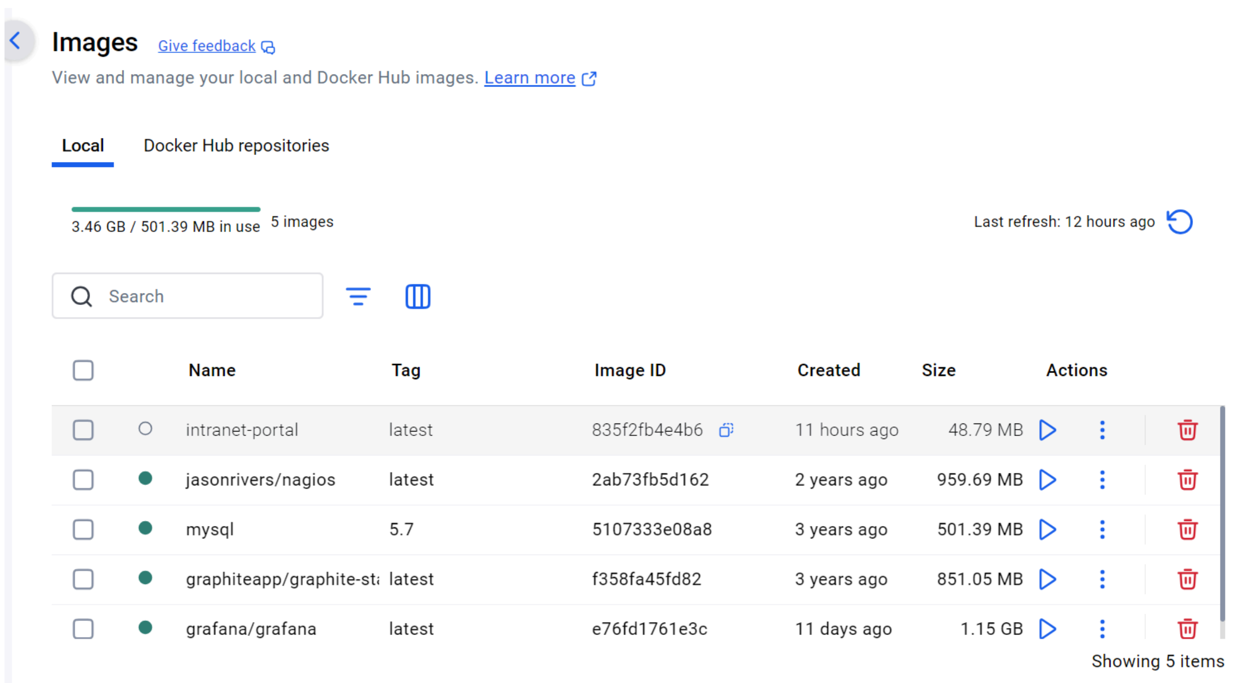
```
CONTAINER ID   IMAGE   COMMAND   CREATED   STATUS   PORTS   NAMES
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker compose up -d
[+] Running 18/18
 ✓ grafana Pulled                                                                                   166.9s
 ✓ graphite Pulled                                                                                  104.4s







[+] Running 7/7
 ✓ Network 24bce2055_jahnavisingh_devops_project_default           Created                             0.1s
 ✓ Volume "24bce2055_jahnavisingh_devops_project_nagios-state"     Created                             0.0s
 ✓ Volume "24bce2055_jahnavisingh_devops_project_graphite-storage" Created                             0.0s
 ✓ Volume "24bce2055_jahnavisingh_devops_project_grafana-storage"  Created                             0.0s
 ✓ Container nagios                                                Started                             1.2s
 ✓ Container graphite                                              Started                             1.3s
 ✓ Container grafana                                               Started                             1.5s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

```
intranet-container
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker compose up -d
[+] Running 32/49
 - grafana [⣿⣿⣀          ] Pulling                                                                   41.2s
 - graphite [      ] Pulling                                                                          41.2s
 - nagios [⣿⣀⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿] 176.4MB / 328.5MB Pulling                                 41.2s
```

```
 => => naming to docker.io/library/intranet-portal:latest                              0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker build -t intranet-portal:latest .
[+] Building 2.1s (14/14) FINISHED                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                0.0s
 => => transferring dockerfile: 365B                                                                0.0s
 => [internal] load metadata for docker.io/library/nginx:1.27-alpine                                1.8s
 => [internal] load metadata for docker.io/library/node:20-alpine                                   1.8s
 => [internal] load .dockerignore                                                                   0.0s
 => => transferring context: 176B                                                                   0.0s
 => [build 1/6] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6  0.0s
 => [internal] load build context                                                                   0.0s
 => => transferring context: 1.03kB                                                                 0.0s
 => [stage-1 1/2] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f  0.0s
 => CACHED [build 2/6] WORKDIR /app/intranet-portal                                                 0.0s
 => CACHED [build 3/6] COPY intranet-portal/package*.json ./                                        0.0s
 => CACHED [build 4/6] RUN npm install                                                              0.0s
 => CACHED [build 5/6] COPY intranet-portal/ ./                                                     0.0s
 => CACHED [build 6/6] RUN npm run build                                                            0.0s
 => CACHED [stage-1 2/2] COPY --from=build /app/intranet-portal/build /usr/share/nginx/html         0.0s
 => exporting to image                                                                              0.0s
 => => exporting layers                                                                             0.0s
 => => writing image sha256:835f2fb4e4b67874b746689b44ada9b9ae5b3a43dac03e88b7f381e054128e44        0.0s
 => => naming to docker.io/library/intranet-portal:latest                                           0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

```
 => => naming to docker.io/library/intranet-portal:latest                                          0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker run -d -p 8080:80 --name intranet-container intr
anet-portal:latest
a68da5bc3ac5572e3be0181a2e68f0fc7f7cdec06cd39b837f27f7e1db460d19
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS         PORTS
 NAMES
a68da5bc3ac5   intranet-portal:latest   "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes   0.0.0.0:8080->80/tcp
 intranet-container
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

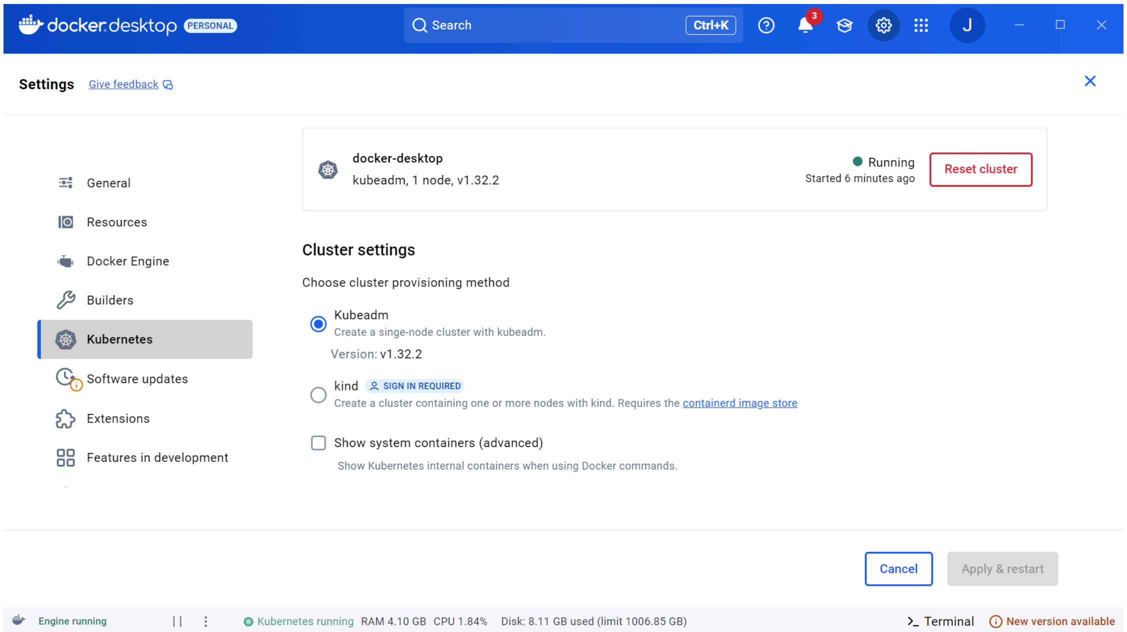




5. Kubernetes Deployment

The portal was deployed using a Kubernetes cluster to manage service lifecycle and scalability. The deployment consists of Pods and Services configured to expose the application to the local network.

- **Service Port:** 30080
- **Status:** All pods were verified to be in the Running state via `kubectl get pods`.




```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl version --client
Client Version: v1.32.2
Kustomize Version: v5.5.0
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl apply -f deployment.yaml
deployment.apps/intranet-portal created
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl apply -f service.yaml
service/intranet-portal-service created
```

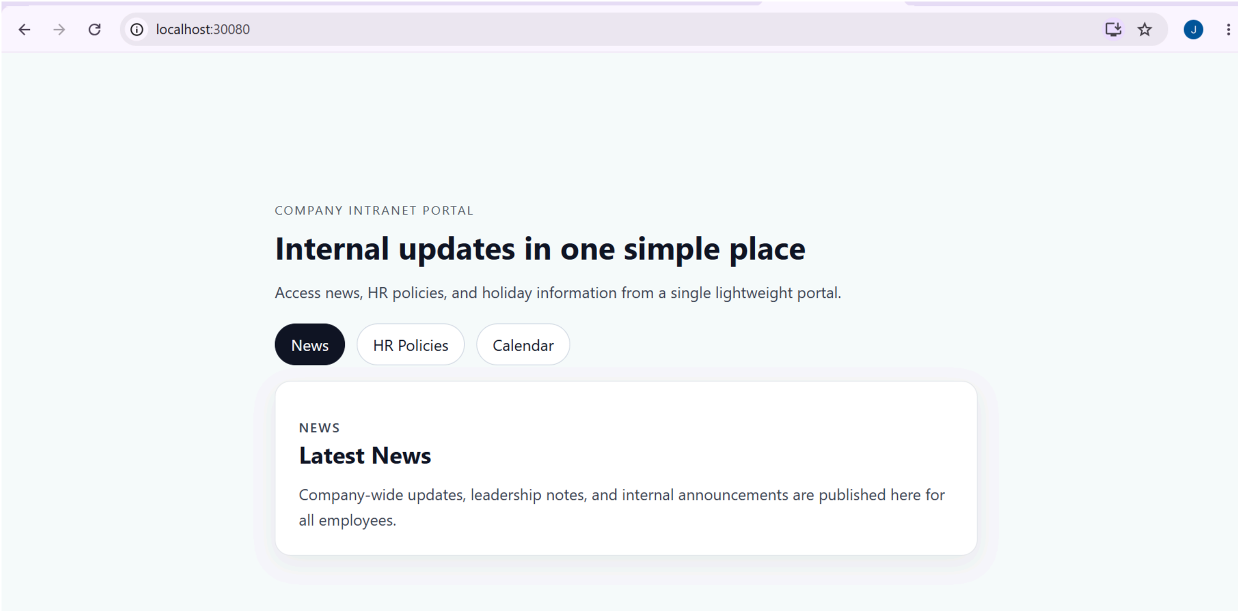
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
intranet-portal-747bd9f899-n5vwp   1/1     Running   0          2m28s
intranet-portal-747bd9f899-rfx7n   1/1     Running   0          2m40s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

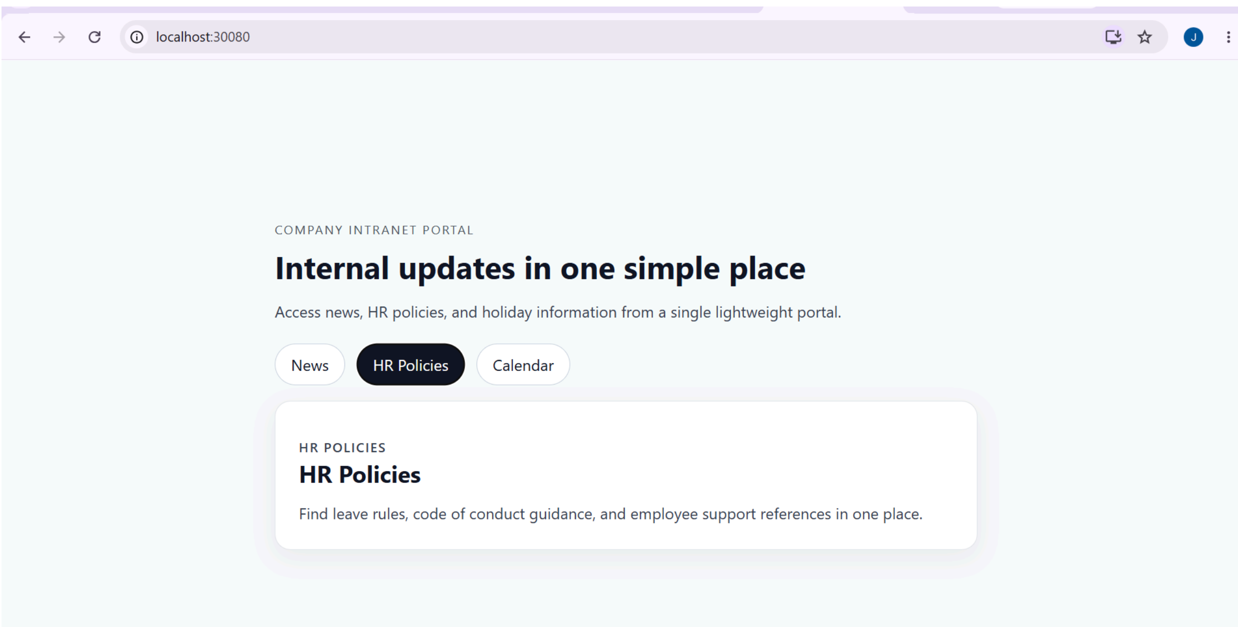
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl get svc
NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
intranet-portal-service   NodePort    10.109.240.99   <none>        80:30080/TCP   22m
kubernetes                ClusterIP   10.96.0.1       <none>        443/TCP        28m
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

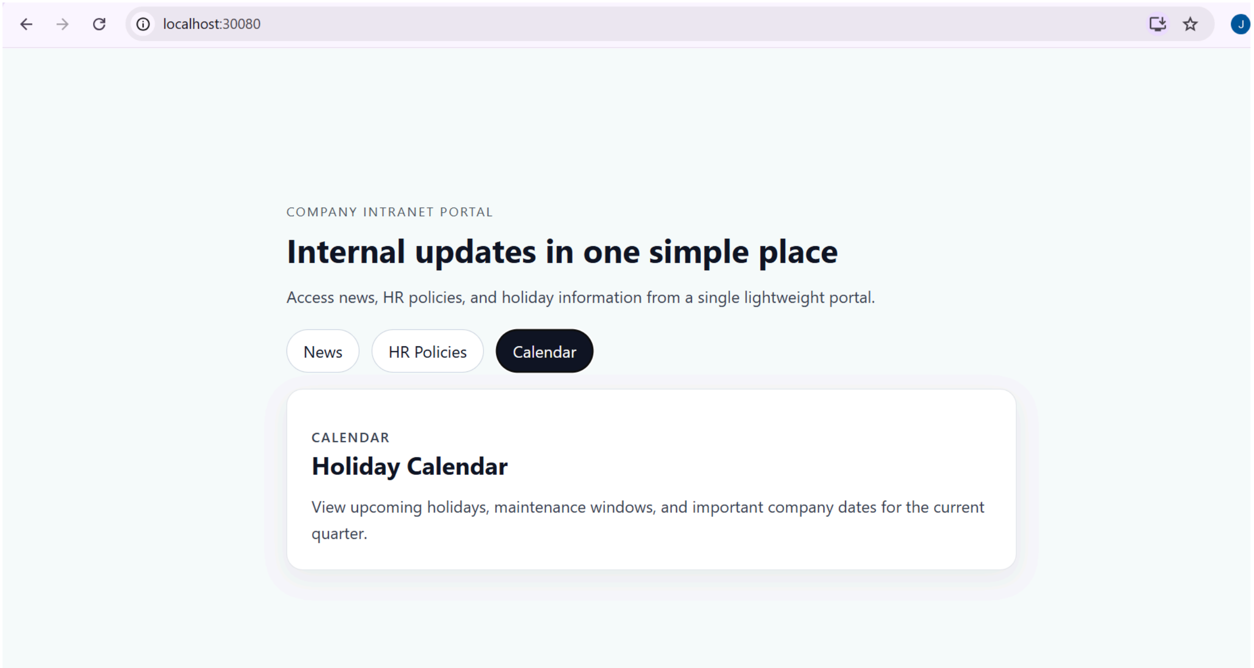
6. Application Output

The application is successfully deployed and reachable.

- **Access URL:** http://localhost:30080


localhost:30080
COMPANY INTRANET PORTAL
Internal updates in one simple place
Access news, HR policies, and holiday information from a single lightweight portal.
News
HR Policies
Calendar
NEWS
Latest News
Company-wide updates, leadership notes, and internal announcements are published here for all employees.

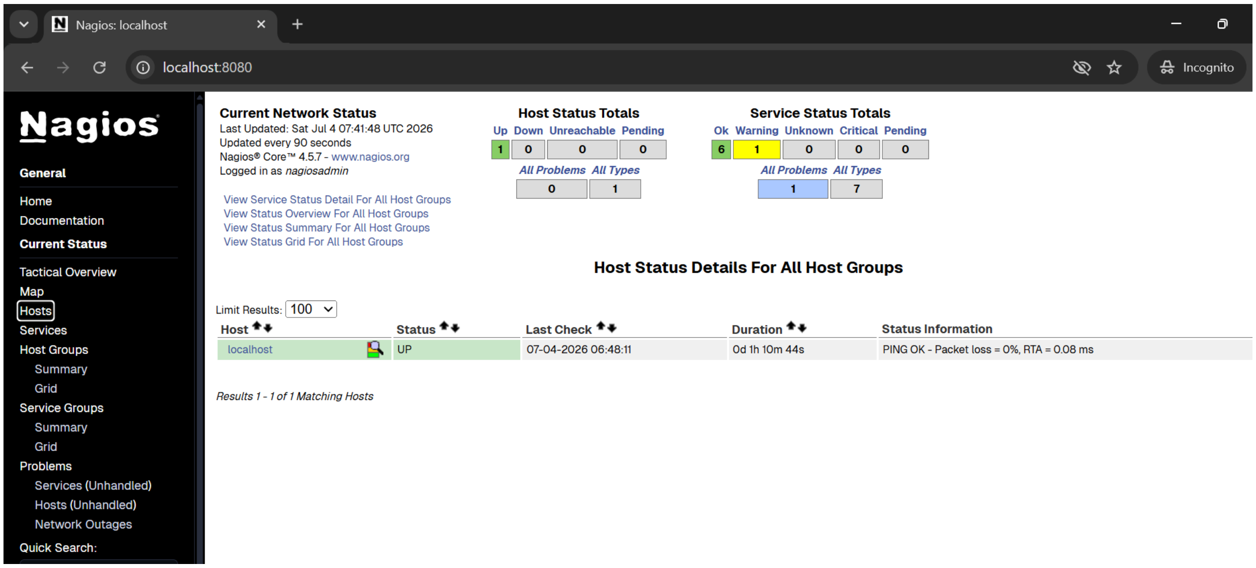

localhost:30080
COMPANY INTRANET PORTAL
Internal updates in one simple place
Access news, HR policies, and holiday information from a single lightweight portal.
News
HR Policies
Calendar
HR POLICIES
HR Policies
Find leave rules, code of conduct guidance, and employee support references in one place.

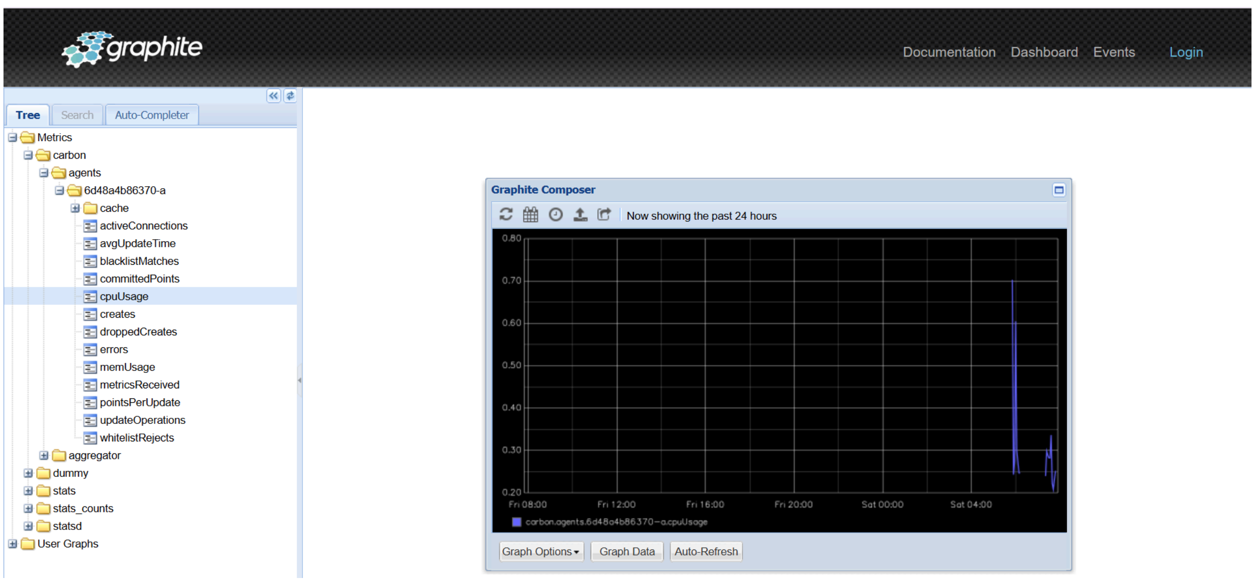


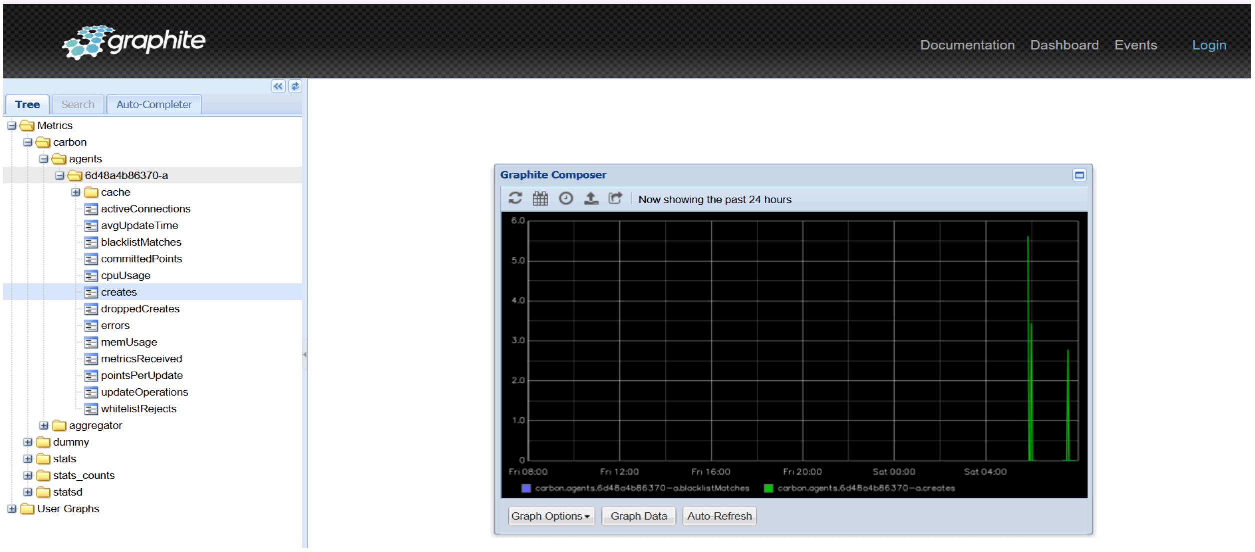
7. Monitoring and Metrics

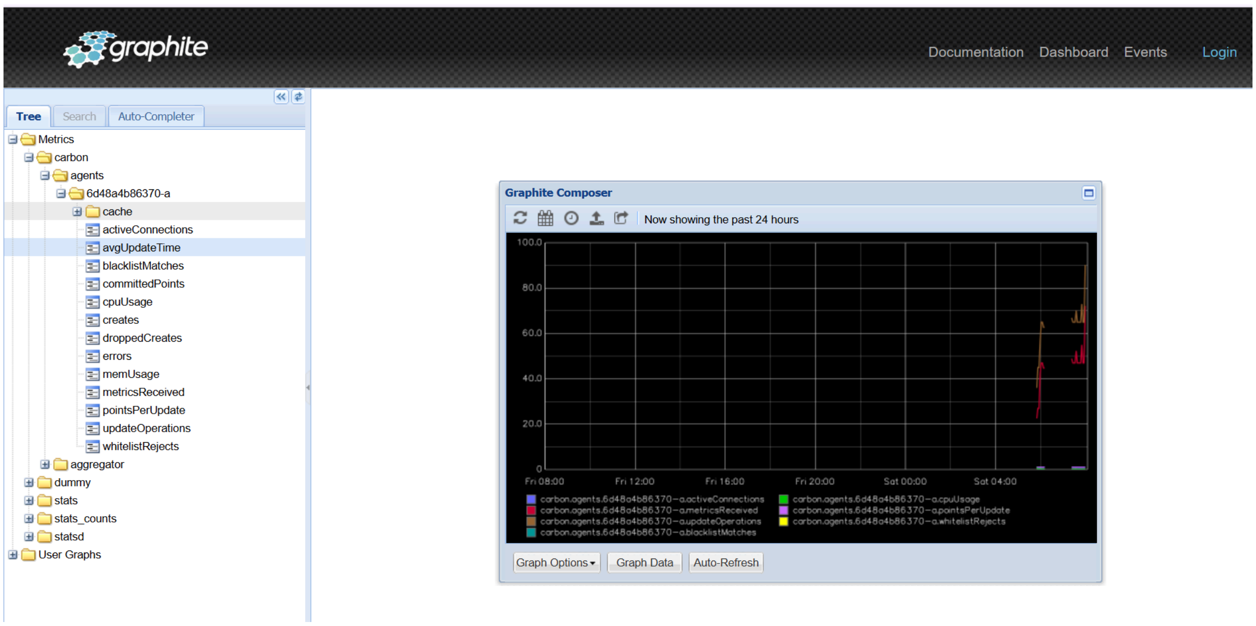
The monitoring stack was configured to provide real time visibility into system performance using the following ports:

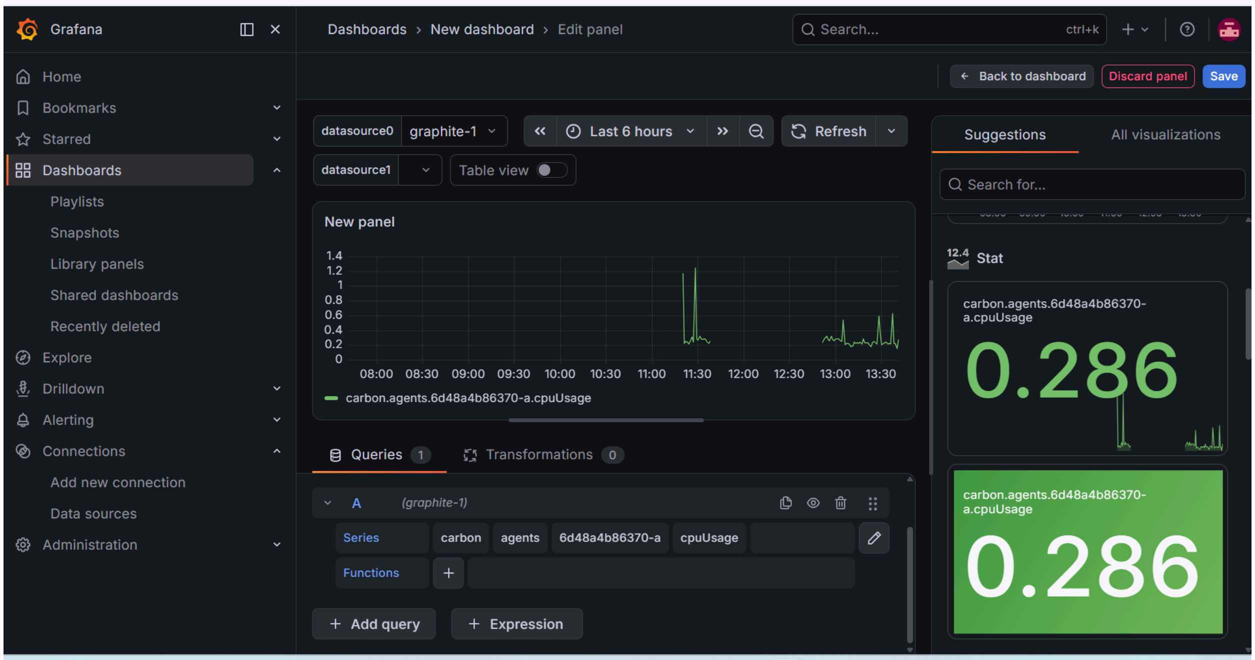
- **Nagios (Port 8080):** This service acts as the primary infrastructure monitor, performing continuous health checks on the system. It verifies connectivity and service responsiveness, currently reporting the host as UP and confirming all critical services are marked as OK.
- **Graphite (Port 8081):** Serving as the core metrics backend, Graphite is responsible for the ingestion and long term storage of time series data. It continuously collects data streams emitted by the application, providing a reliable foundation for performance analysis.
- **Grafana (Port 3001):** Grafana functions as the visualization layer, connected directly to the Graphite data source. It transforms raw metric data into intuitive dashboards that display key performance indicators such as CPU utilization, memory consumption, and request latency, enabling proactive system management.

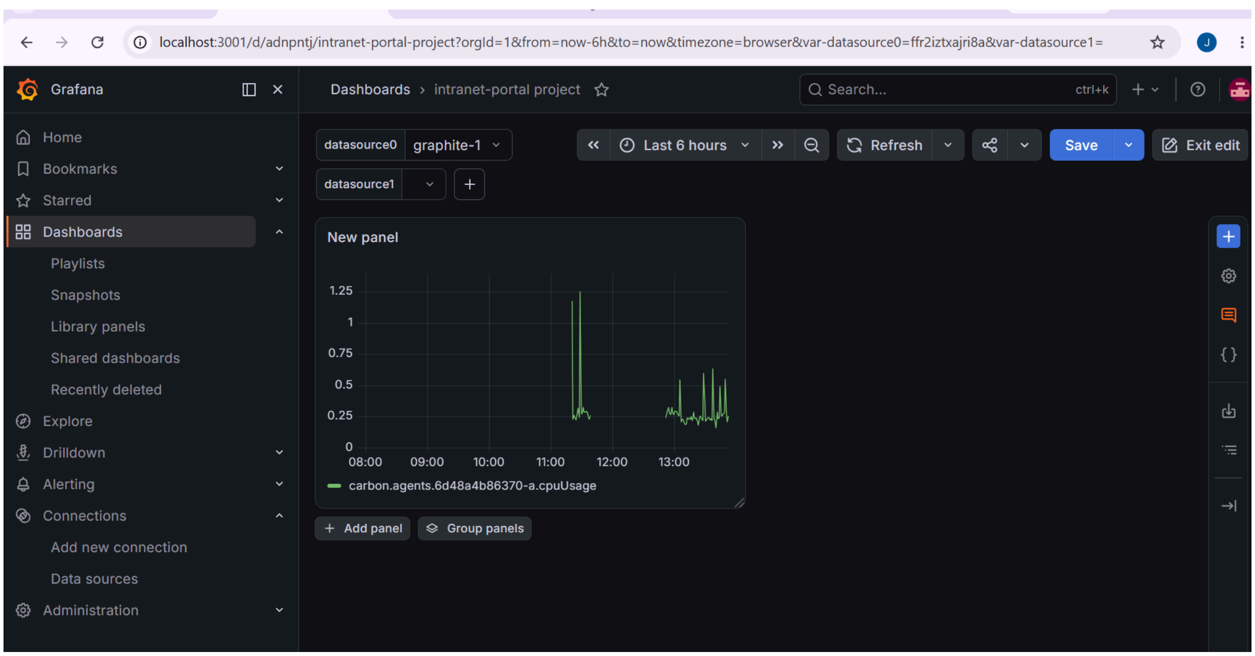

Nagios: localhost
localhost:8080
Incognito
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Current Network Status
Last Updated: Sat Jul 4 07:41:48 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.7 - www.nagios.org
Logged in as nagiosadmin
View Service Status Detail For All Host Groups
View Status Overview For All Host Groups
View Status Summary For All Host Groups
View Status Grid For All Host Groups
Host Status Totals
Up Down Unreachable Pending
1 0 0 0
All Problems All Types
0 1
Service Status Totals
Ok Warning Unknown Critical Pending
6 1 0 0 0
All Problems All Types
1 7
Host Status Details For All Host Groups
Limit Results: 100
Host Status Last Check Duration Status Information
localhost UP 07-04-2026 06:48:11 0d 1h 10m 44s PING OK - Packet loss = 0%, RTA = 0.08 ms
Results 1 - 1 of 1 Matching Hosts


graphite
Documentation Dashboard Events Login
Tree Search Auto-Completer
Metrics
carbon
agents
6d48a4b86370-a
cache
activeConnections
avgUpdateTime
blacklistMatches
committedPoints
cpuUsage
creates
droppedCreates
errors
memUsage
metricsReceived
pointsPerUpdate
updateOperations
whitelistRejects
aggregator
dummy
stats
stats_counts
statsd
User Graphs
Graphite Composer
Now showing the past 24 hours
0.80
0.70
0.60
0.50
0.40
0.30
0.20
Fri 08:00
Fri 12:00
Fri 16:00
Fri 20:00
Sat 00:00
Sat 04:00
carbon.agents.6d48a4b86370-a.cpuUsage
Graph Options
Graph Data
Auto-Refresh


graphite
Documentation
Dashboard
Events
Login
Tree
Search
Auto-Completer
Metrics
carbon
agents
6d48a4b86370-a
cache
activeConnections
avgUpdateTime
blacklistMatches
committedPoints
cpuUsage
creates
droppedCreates
errors
memUsage
metricsReceived
pointsPerUpdate
updateOperations
whitelistRejects
aggregator
dummy
stats
stats_counts
statsd
User Graphs
Graphite Composer
Now showing the past 24 hours
Graph Options
Graph Data
Auto-Refresh


graphite
Documentation
Dashboard
Events
Login
Tree
Search
Auto-Completer
Metrics
carbon
agents
6d48a4b86370-a
cache
activeConnections
avgUpdateTime
blacklistMatches
committedPoints
cpuUsage
creates
droppedCreates
errors
memUsage
metricsReceived
pointsPerUpdate
updateOperations
whitelistRejects
aggregator
dummy
stats
stats_counts
statsd
User Graphs
Graphite Composer
Now showing the past 24 hours
Graph Options
Graph Data
Auto-Refresh


Grafana
Dashboards › New dashboard › Edit panel
Search... ctrl+k
Back to dashboard
Discard panel
Save
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
datasource0 graphite-1
Last 6 hours
Refresh
datasource1
Table view
New panel
1.4
1.2
1
0.8
0.6
0.4
0.2
0
08:00 08:30 09:00 09:30 10:00 10:30 11:00 11:30 12:00 12:30 13:00 13:30
carbon.agents.6d48a4b86370-a.cpuUsage
Queries 1
Transformations 0
A (graphite-1)
Series carbon agents 6d48a4b86370-a cpuUsage
Functions
Add query
Expression
Suggestions
All visualizations
Search for...
Stat
carbon.agents.6d48a4b86370-a.cpuUsage
0.286
carbon.agents.6d48a4b86370-a.cpuUsage
0.286


localhost:3001/d/adnpntj/intranet-portal-project?orgId=1&from=now-6h&to=now&timezone=browser&var-datasource0=ffr2iztxajri8a&var-datasource1=
Grafana
Dashboards › intranet-portal project
Search... ctrl+k
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
datasource0 graphite-1
Last 6 hours
Refresh
Save
Exit edit
datasource1
New panel
1.25
1
0.75
0.5
0.25
0
08:00 09:00 10:00 11:00 12:00 13:00
carbon.agents.6d48a4b86370-a.cpuUsage
Add panel
Group panels